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Spatially dependent polarization of PVDF nanocomposites under the combined influence of an electric field and interfacial interaction

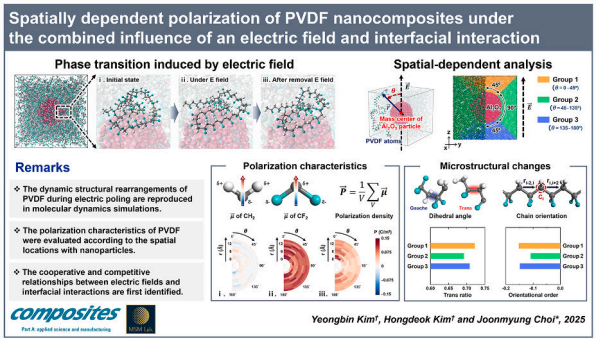
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HIGHLIGHTS

- The polarization behavior of PVDF/Al_2O_3 nanocomposites is spatially evaluated.
- The electric poling of PVDF is described by repeated electric field application.
- Polarization is governed by the combination of electric field and hetero-interaction.
- The electric field induces a permanent and spatial ly independent PVDF polarization.
- Polarization locking using nanoparticles exhibits spatial selectivity.

GRAPHICAL ABSTRACT



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ABSTRACT

Highly polarized nanoparticles enhance the piezoelectric and mechanical properties of polyvinylidene fluoride (PVDF) by promoting the formation of its electroactive β-phase during electric poling. In this study, the polarization-formation mechanism of PVDF/Al_2O_3 nanocomposites under repeated electric field application was investigated using molecular dynamics simulations. The results revealed that the combined effects of the electric field and interfacial interactions led to spatially dependent polarization characteristics. The electric field induced unidirectional transformations of the PVDF chains, resulting in permanent and spatially independent polarization changes. By contrast, the polarization-locking effect induced by the nanoparticles varied significantly based on the relative positions of the nanoparticles. Thus, the residual polarization level and the associated structural transformations of the PVDF chains were determined by combining the electric field and nanoparticle adsorption. A detailed analysis of the dihedral angle distributions and orientational orders successfully revealed the molecular-level origin of these spatial differences in the polarization characteristics. This study offers methodological guidelines for the rational design of nanocomposites with enhanced performance by controlling synergistic nanoparticle interactions.

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1. Introduction

Polyvinylidene fluoride (PVDF) is a piezoelectric polymer that converts mechanical deformation into electrical energy [1–3]. As a semi-crystalline polymer, PVDF predominantly exists in three phases, namely, α(TGTG′), β(TTTT), and γ(TTTGTTTG′), depending on the combination of *trans* (T) or *gauche* (G) conformations of the carbon backbone [4,5]. Among them, the β phase consists of all *trans* backbone units. The dipole moments of the PVDF chains are aligned parallel to each other, resulting in a high level of spontaneous polarization [6]. Owing to the strong ferroelectric and piezoelectric properties of the β phase, PVDF can be extensively applied in advanced technologies, such as flexible sensors, energy-harvesting systems, wearable electronic devices, and electroactive actuators [7–13].

Consequently, strategies to induce and stabilize the β phase are important for improving the performance and functionality of PVDF-based devices. Recent efforts to enhance the electroactive properties of PVDF have focused on the incorporation of ceramic nanofillers such as Al_2O_3, ZnO, and $BaTiO_3$, followed by electric field poling [14–22]. These polar ceramic fillers act as nucleating agents for β-phase crystals, which significantly enhance the crystallization kinetics and the degree of the polar phase [23–25]. The highly polarized surface of the filler interacts electrostatically with the partially charged PVDF chains, facilitating the β-phase transformation and helping maintain it, even after electric field removal [26]. Furthermore, efficient stress transfer between adjacent nanoparticles under a mechanical load enhances the electrical output of PVDF near the interface [27–31].

The electric poling behavior of PVDF nanocomposite systems is governed by the combined influence of the external electric field and interfacial interactions in the nanofiller. This poses notable challenges at the interfaces with the nanoparticles because the electricity-harvesting efficiency is spatially differentiated within the composite. The isotropic attraction of nanoparticles and the orientation guidance of the electric field are reconciled differently depending on the spatial location, leading to spatially non-uniform polarization characteristics. Although previous studies have highlighted the improved poling efficiency and load-transfer capability of nanocomposite systems, a theoretical framework capable of interpreting the spatially heterogeneous energy-harvesting behavior based on the relative position of nanofillers has not yet been developed [15,32–34]. It remains challenging to experimentally identify how the local PVDF chain configuration near the interface differs from that in the bulk during field-induced β phase formation. From a numerical modeling perspective, it has often been assumed that the PVDF matrix exhibits spatially uniform and homogeneous electromechanical properties regardless of the nanoparticle distribution [35–38]. In other words, the spatial variation in the microstructural rearrangement of PVDF around the nanoparticles and the corresponding macroscopic polarization influence are insufficiently understood.

Motivated by these challenges, this study aims to elucidate the polarization-formation mechanism in PVDF/Al_2O_3 nanocomposites under electric fields and to investigate the spatial dependence of the polarization density. Using molecular dynamics (MD) simulations, the dynamic response of PVDF polymers under repeated electric field loading was continuously monitored. Particular emphasis was placed on the competitive and cooperative relationship between spatial confinement by the nanoparticles and orientation guidance by an electric field. This approach reveals how the anisotropic polarization-locking effect of nanoparticles leads to spatially heterogeneous polarization characteristics. Consequently, this study not only offers a fundamental understanding of the polarization phenomena in PVDF-based nanocomposites, but also provides theoretical guidance for the design of high-performance piezoelectric materials by controlling the nanoparticle geometry and spatial distribution.

2. All-atom modeling of neat PVDF and PVDF/Al_2O_3 nanocomposites

To investigate the effects of the spherical nanoparticle interactions on the phase transition of PVDF during the application of an external electric field, all-atom MD models of neat PVDF and PVDF/Al_2O_3 nanocomposites were constructed. The Al_2O_3 nanofiller considered in this study is an insulating material, which has the advantage of allowing understanding of the effect of interfacial interactions at a purely structural level. Other materials (e.g., semiconductors and ferroelectric ceramics) show dramatic changes in their surface polarization properties under electric fields [39,40]. In this respect, the insulating properties of Al_2O_3 are suitable for the subject of this study as they ensure high preservation of the initial structure even in very high electric field environments. This also simplified the β-phase formation process of PVDF chains to a problem that only depends on the physical adsorption characteristics at the interface by nanoparticles and the direction of the electric field.

Fig. 1**(a)** shows the single PVDF chain and spherical Al_2O_3 nanoparticles used in this study. Twelve PVDF chains consisting of 240 repeating units (molecular weight of 15388.4 g/mol) were randomly distributed in a periodically bounded cubic unit cell. To avoid unintended interference between molecules during positioning, the unit cells were initially constructed at a low density of 0.4 g/cc. Structural equilibrium was achieved using the conjugate gradient method with a cutoff energy of 0.0001 kcal/mol. Subsequently, the unit cell was spontaneously compressed toward equilibrium under the NPT ensemble at 300 K and 1 atm for 1 ns. The equations of motion were integrated using the Verlet algorithm with a time step of 1 fs. The cutoff distance for non-bonded interactions was set to 10 Å, and coulombic forces were computed using the Ewald summation method with an accuracy of 0.0001 kcal/mol. The nanocomposite model was constructed in the same manner, except that a single nanoparticle with a diameter of 3 nm was integrated at the center of the unit cell. The corresponding volume fraction of the nanoparticle was 7.8 vol%. Finally, the microstructures of both models were further equilibrated for 2 ns under an NPT ensemble at 300 K and 1 atm. The configurations of the completed models are shown in Fig. 1**(b)**.

The phase-transition behavior of the PVDF chains was analyzed by repeatedly applying an electric field to the two prepared models. A strong electric field of 5 GV/m was applied in the z-direction for 0.5 ns, following which time integration was applied for an additional 0.5 ns in an environment where the electric field was released. The application and release of the electric field were repeated 10 times, and the PVDF chain structure changed cumulatively during the process. After completion of the final electric field application cycle, an additional equilibration period of 4 ns was provided. Note that the high-intensity electric fields provided in the present simulations did not induce any response other than accelerating the structural orientation of the polarizable material. Even when high-level electric fields are applied, the temperature and pressure environment of the entire simulation box successfully maintains equilibrium (see **Fig. S1**). Also, the covalent bond between atoms under applied conditions is described as an unbreakable harmonic spring, so unintended dissociation and breakage of covalent bonds does not occur within this framework. Such an approach enables successful tracking of the phase transition trends of PVDF under an all-atom MD simulation environment [15,32,41,42].

As the electric field was applied at each cycle, PVDF underwent a β-phase transition, which resulted in a change in the dihedral angle of the backbone [43]. As shown in Fig. 1**(c)**, the applied external energy causes the dihedral angles of the PVDF backbone to shift from a stable *gauche* structure (0°–120°, 240°–360°) to a metastable *trans* structure (120°–240°). From a macromolecular point of view, such an increase in *trans* states can be understood as an increase in the probability of existence of the β-phase in the microstructure [42]. To quantify these transition characteristics, the *trans* ratio was calculated as the number of

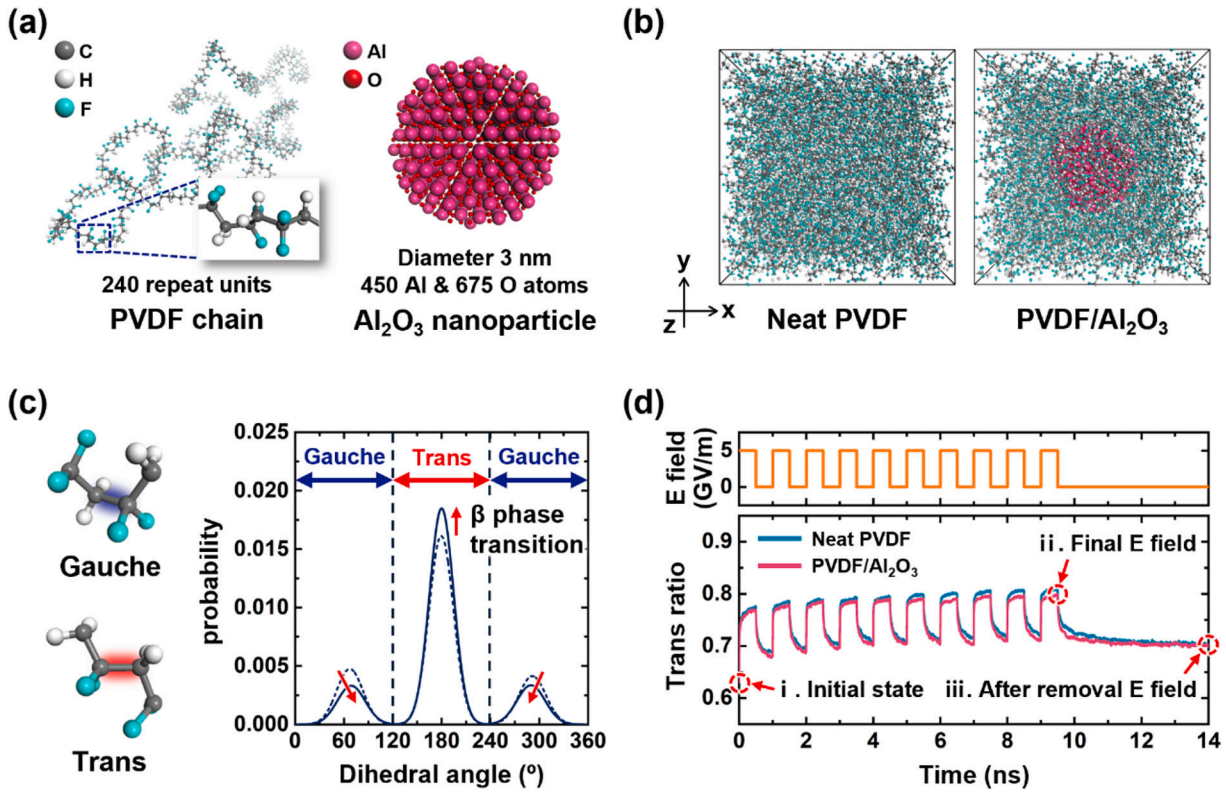


Fig. 1. Phase-transition simulation of PVDF under electric fields. (a) Molecular structures of a linear PVDF chain (left) and spherical Al_2O_3 nanoparticle (right). (b) All-atom configuration of equilibrated PVDF and PVDF/Al_2O_3 nanocomposite microstructures. (c) Probability distribution of the dihedral angles of the PVDF backbone. The distribution was obtained by fitting the data over the entire microstructure to a Gaussian function. The dihedral angles are classified as either *gauche* (blue) or *trans* (red), based on their respective ranges. The dotted and solid line profiles represent the dihedral angle distributions before and after the application of the electric field, respectively. (d) Changes in the *trans* backbone ratio under periodic electric field applications.

dihedral angles in the *trans* structure with respect to the total number of dihedral angles in the PVDF chain (Fig. 1**(d)**). The electric field increased the formation ratio of the *trans*-PVDF backbone by more than 26 %. After releasing the electric field, some springback occurred; nevertheless, the final *trans*-state ratio increased by ~ 11 % compared with that of the initial state. Such a phase transition rate is the result of the combined effect of PVDF chains wrapping the nanoparticles by physical adsorption around Al_2O_3 nanoparticles with a diameter of 3 nm and the directionality induced by the electric field. Meanwhile, such an increase in the *trans* ratio was observed gradually under repeated electric fields, indicating that the β-phase transition level of PVDF can be precisely controlled by the user.

We selected three representative states based on the distinct structural changes that the model experiences during the aforementioned process: (i) a fully equilibrated state before the application of an electric field, (ii) immediately before the removal of the electric field in the final cycle, and (iii) after stabilization following periodic electric field application. The structural changes in the PVDF chains and the interactions between the spherical nanoparticles and polymer structure were thoroughly examined and are discussed in the following sections.

We performed a thorough comparative analysis between our simulation results before and after repeated electric field and relevant experimental data reported in the literature (see **Table S1** and **Table S2**) [S1-S7]. The detailed process for calculating the properties is summarized in **Supplementary Text 1**. The material composition, structural, and thermomechanical properties of the PVDF model prior to electric poling were quantitatively predicted in satisfactory agreement with experimentally reported values. In addition, the polarization characteristics and thermomechanical properties were significantly improved after repeated electric field application, which also coincides well with experimental observations. Such improved properties were mainly attributed to structural rearrangement into the β-phase induced by electric poling. The calculated polarization densities also fall within the experimentally observed range. We also noted changes in film transmittance that were confirmed from an optical perspective. It is known that repeated application of an electric field increases the opacity of the entire film, which is due to an increase in the β-phase abundance of PVDF. This is qualitatively consistent with the results of our simulations, where the *trans*-structure fraction increases as PVDF forms a highly polarized phase. The above observations demonstrate that the microstructural changes, such as dihedral angle changes and chain orientations induced by repetitive electric poling, successfully explain the experimentally verified properties.

The other simulation details are as follows. The initial model was constructed using a commercial MD simulation package, Materials Studio 2020 (Dassault Systèmes, France). Electric field applications and associated thermodynamic processes were performed using the open-source code LAMMPS (Sandia Lab., USA). The atomic interactions within the PVDF polymer were described using a polymer consistent force field (PCFF) [44]. The ionic interactions within the Al_2O_3 nanoparticles were modeled using the Buckingham potential [45–48].

3. Results and discussion

3.1. Characteristics of interactions between Al_2O_3 nanoparticles and PVDF chains

To better understand the electric field-induced structural changes in the PVDF chains interacting with Al_2O_3, we first analyzed the interfacial interaction characteristics in the initial state without an applied electric field (i.e., stage (i) in Fig. 1**(d)**). The components of Al_2O_3 are coupled by strong ionic interatomic bonding, with the Al atoms carrying a positive

charge (+1.8) and O atoms carrying a negative charge (–1.2). The PVDF chains also possess partial charges; hence, the electrostatic potential plays an important role in determining the structural properties of the polymer chains around the Al_2O_3 surface. Fig. 2**(a)** shows the molecular arrangement of PVDF in the interfacial region at equilibrium. The PVDF chains are wrapped around the Al_2O_3 nanoparticles along the surface curvature through electrostatic interactions. No significant preference is observed for the type of terminal group (CH_2 or CF_2) of the PVDF chains on the nanoparticles. Such atomic species-specific distributions were confirmed by the radial distribution functions (RDFs) of Al_2O_3–H and Al_2O_3–F. As shown in Fig. 2**(b)**, no significant selective absorption of H and F atoms is observed on the nanoparticle surface. In other words, the PVDF chains largely preserve their random and amorphous arrangement in the interfacial region, even while interacting with Al_2O_3.

Interestingly, despite the similarities in the positional states of hydrogen and fluorine, the influence of their interfacial energies significantly differed. This is clearly revealed when the RDF profiles of the Al and O atoms constituting the nanoparticles and the H and F atoms constituting the PVDF chains are derived separately, as shown in Fig. 2 **(c)**. The results show that the Al atoms interact more strongly with F atoms, whereas the O atoms interact more strongly with H atoms. Such component-dependent properties are plausible when considering the partially charged states formed by the nanoparticles and polymer chains. Notably, the charge difference between Al and F is much more

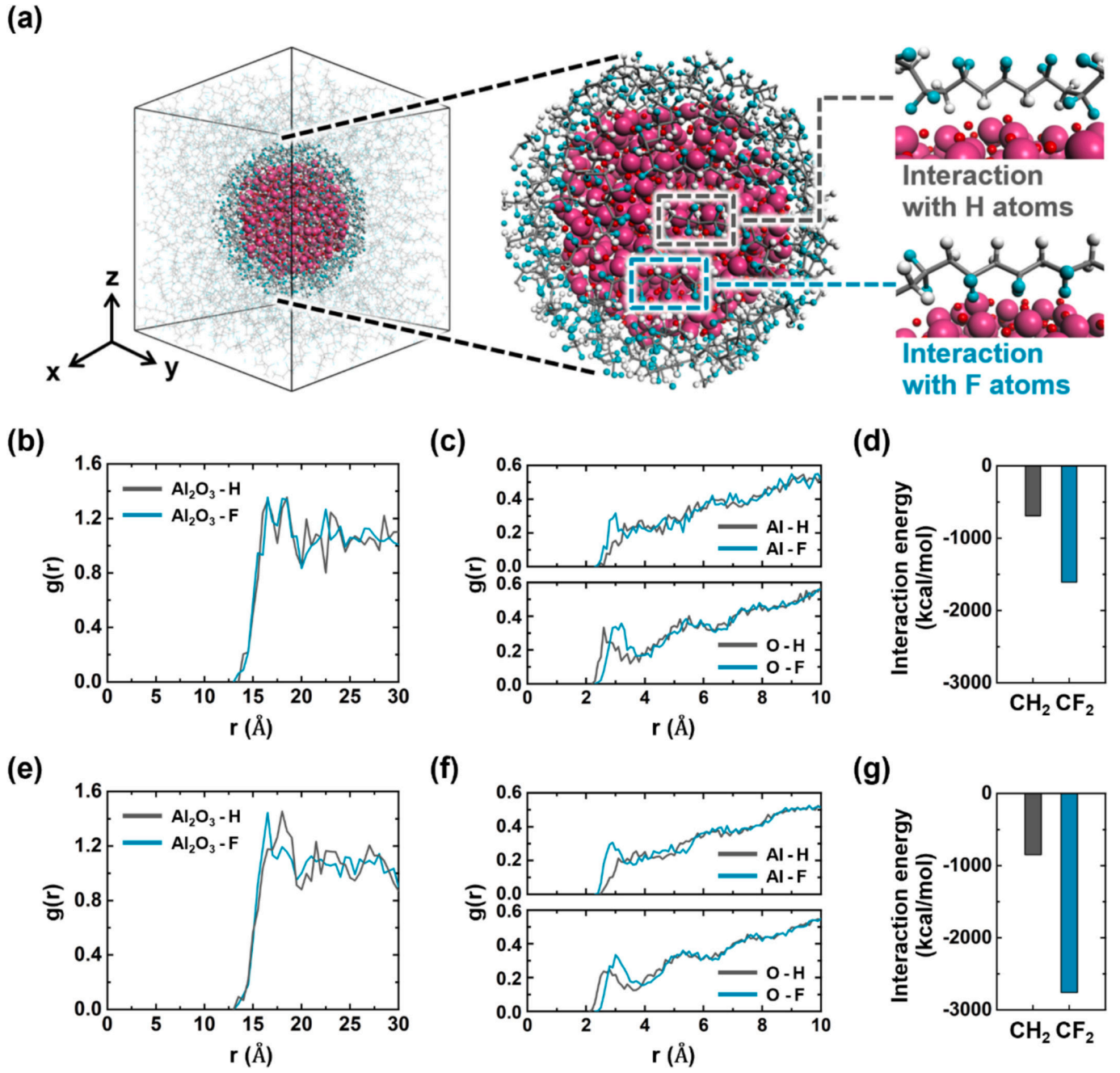


Fig. 2. Molecular-level characterization of interaction behaviors at the nanocomposite interface. (a) Schematic of the interaction between spherical Al_2O_3 nanoparticle and PVDF chains. (b–d) Interfacial structure and interaction profiles before electric field application: (b) spatial distribution of H and F atoms of PVDF relative to the Al_2O_3 surface; (c) component-wise subdivision of radial distribution function (RDF) profiles; (d) comparison of interfacial interaction energies between nanoparticle–CH_2 and nanoparticle–CF_2 groups. (e–g) Analysis of the counterpart after electric field application: (e) spatial distribution of H and F atoms; (f) subdivided RDF profiles; (g) interaction energies with the nanoparticle.

pronounced than that between O and H. This indicates that although the terminal groups of the PVDF chains appear to be randomly arranged around the nanoparticles, their contributions to the interfacial energy formed by each group are vastly different. Indeed, when the interfacial interaction energies of CH_2 and CF_2 are separately evaluated, as shown in Fig. 2**(d)**, the CF_2 groups exhibit a much stronger interaction energy than the CH_2 groups. These differences in the interaction energy are the origin of the programming of the PVDF chains around the nanoparticles into specific arrangements under an electric field; this aspect is discussed in detail in the following sections.

Another interesting aspect is the change in behavior of the polymer in the interfacial region after an electric field is applied. After the PVDF/Al_2O_3 nanocomposite model was re-equilibrated under an electric field, the structural properties of the PVDF chains located in the interfacial region were reanalyzed (Fig. 2**(e-g)**). As presented in Fig. 2**(e)**, the first peak of F atoms lies closer to the nanoparticle surface than that of H atoms. This is completely different from the initial state where no significant difference was observed in the first peak positions of H and F. In contrast, the component-specific RDF profiles of Al–F, Al–H, O–F, and O–H pairs exhibit minimal changes (Fig. 2**(f)**), suggesting that the fundamental properties of nanoparticle-polymer interactions remain consistent under the electric field. Moreover, the interfacial interaction energy between the CF_2 groups and the nanoparticles significantly increased after electric polarization as shown in Fig. 2**(g)**. This confirms that the electric field-induced alignment effectively fixes the polarization, mainly by enhancing the interaction through the CF_2 groups.

3.2. Spatially dependent polarization characteristics of PVDF molecules around Al_2O_3 nanoparticles

The polarization characteristics and structural changes in the PVDF chains under the applied electric field were analyzed based on their relative positions around the spherical nanoparticles. Fig. 3**(a)** shows this classification based on the offset angle, θ, which is defined as the angle between the electric field direction vector (i.e., the + z-axis) and the vector extending from the mass center of a nanoparticle to a specific atom in a PVDF chain. The θ values were calculated for all the atoms comprising PVDF, and were then grouped into three regions, as illustrated in Fig. 3**(b)**. Group (1) is the region covering the surface of the Al_2O_3 nanoparticles, where θ is 0°–45°. For the atoms in this region, both the electric field and interaction force with Al_2O_3 are parallel to the z-axis and have the same direction. Group (2) is the region in which θ is in the range of 45°–135°, where the electric field and the interaction with Al_2O_3 are perpendicular to each other. Group (3) is the region where θ is in the range of 135°–180°; here, both forces are parallel to the z-axis, but their directions are opposite. This grouping facilitates an understanding of the spatial location-dependent behavior of PVDF chains exposed to an electric field and adsorbed to nanoparticles.

When an external electric field is applied, the adsorption effect of Al_2O_3 nanoparticles, along with the structural rearrangement effect of the PVDF chains owing to the polarization of the fluorine-based functional groups themselves, act together. The polarization density in the matrix region under the application of an electric field was calculated based on the dipole moments of all PVDF chains. The dipole moment of each CH_2 and CF_2 group in the PVDF chains was determined based on the following equations:

$$\vec{\mu}_{CH_2} = q_C \vec{r}_C + q_{H1} \vec{r}_{H1} + q_{H2} \vec{r}_{H2} \quad (1)$$

$$\vec{\mu}_{CF_2} = q_C \vec{r}_C + q_{F1} \vec{r}_{F1} + q_{F2} \vec{r}_{F2} \quad (2)$$

where q represents the charge of each atom in the terminal group and $\vec{r}$ is the position vector of the corresponding atom. The dipole moments calculated for each atom were summed separately, depending on whether they belonged to the CH_2 or CF_2 group (see **Fig. S2**). The polarization density, $\vec{P}$, for each measured volume was calculated as follows:

$$\vec{P} = \frac{1}{V}\sum_i^N \vec{\mu}_i \quad (3)$$

where V denotes the volume of the analyzed region and N represents the number of dipole moments contained in that volume. The polarization density was analyzed along the z-axis, corresponding to the direction of the applied electric field. Color maps of the polarization density within the PVDF matrix domains were computed at radial intervals of 1.5 Å from the Al_2O_3 particle surface and angular increments of 22.5° (θ) relative to the z-axis.

Fig. 4 illustrates the spatially distinct polarization characteristics of the PVDF matrix under the influence of an external electric field and its interactions with the nanoparticles. The analyses were performed at three independent time points: stages i, ii, and iii, as defined in Fig. 1**(d)**. Initially, the PVDF matrix exhibited neutral polarization at all spatial locations. However, because the electric field provided a guiding direction for the spontaneous alignment of the PVDF molecules, the polarization density was significantly enhanced. Upon the removal of the electric field, the PVDF molecules sprung back to their initial structure; however, a significant polarization density remained owing to the permanent structural change [41,49].

More interesting characteristics were exhibited by each spatial region, defined as Groups (1), 2, and 3 in Fig. 3. After removing the electric field, Group (2) exhibited a 63.7 % decrease in the polarization density, whereas Groups (1) and (3) showed smaller reductions of 45.8 % and 48.4 %, respectively. This behavior indicates that the polarization-locking effect of the nanoparticles operates spatially differently [50,51]. The highly polarized nanoparticle surface allows the chains to

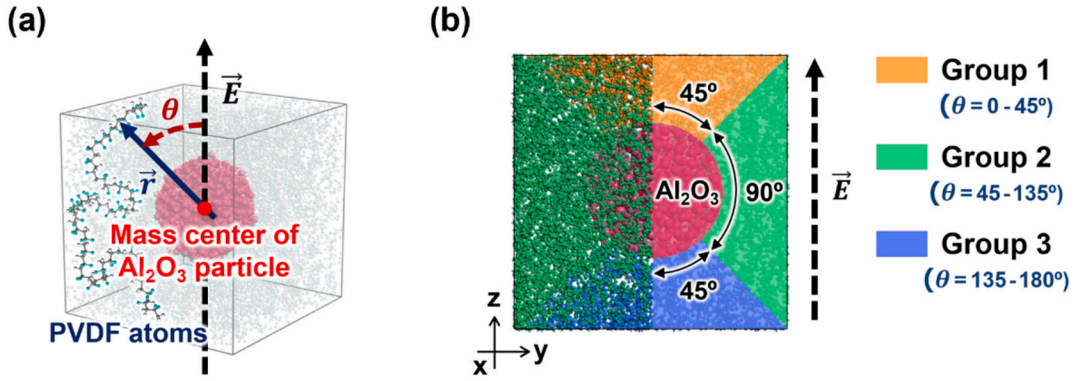


Fig. 3. Analysis of PVDF chain properties based on relative positions to the spherical Al_2O_3 nanoparticle. (a) Schematic of θ. The position vector $\boldsymbol{r}$ represents the relative position from the center of mass of the Al_2O_3 nanoparticle to any atom within the PVDF chain. The offset angle θ is defined as the angle between this position vector $\boldsymbol{r}$ and the electric field direction (i.e., the + z-axis). (b) Classification of the PVDF components into three groups based on θ.

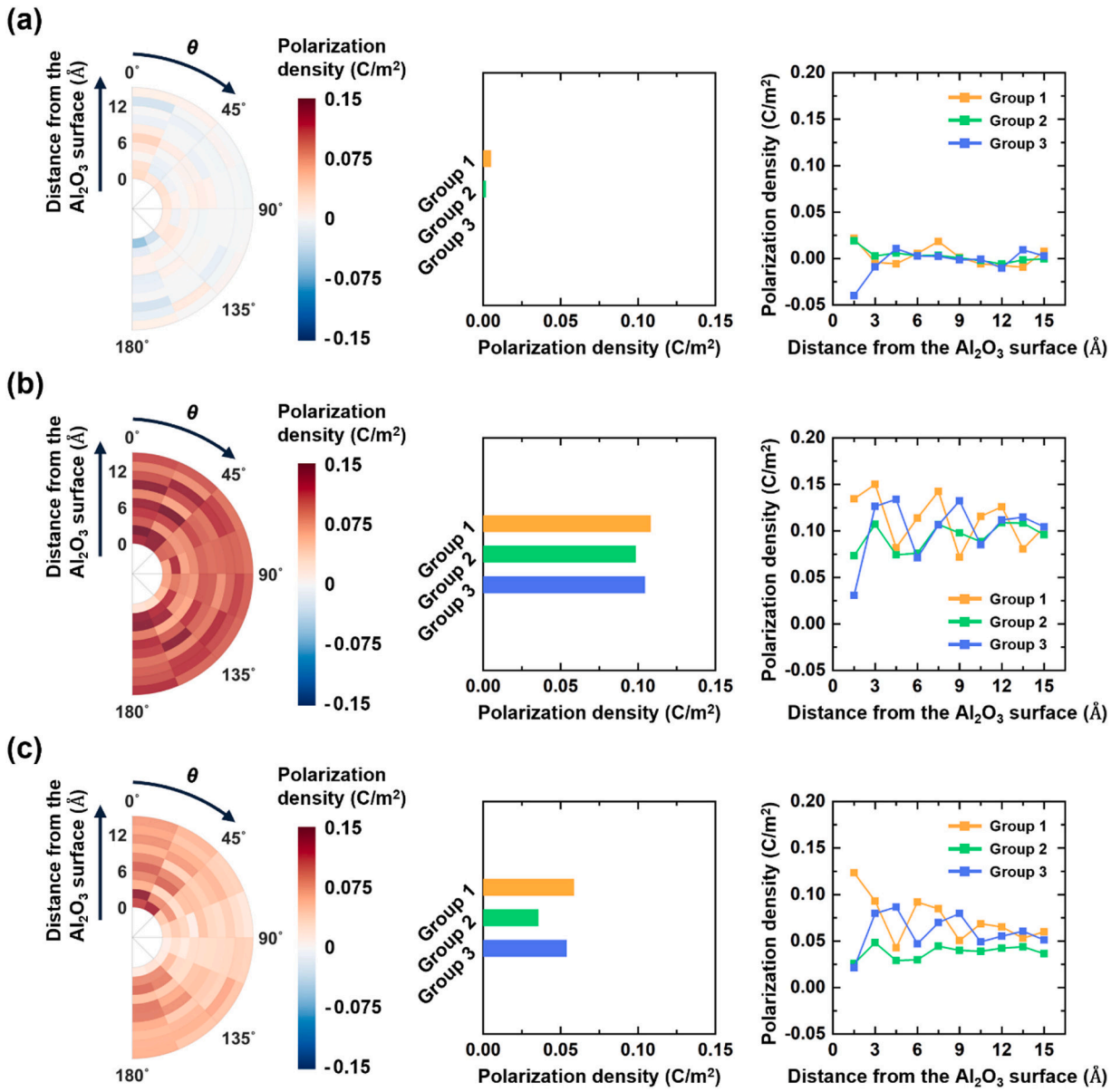


Fig. 4. Polarization characteristics of the PVDF matrix surrounding the spherical Al_2O_3 nanoparticle under an applied electric field. (a) Spatial distribution of the polarization density according to the relative position to the Al_2O_3 nanoparticle (left); the polarization density occupied by each of the regions defined as Groups (1), 2, and 3 (middle); and radial distribution of the polarization density as a function of the distance from the Al_2O_3 nanoparticle surface (right). The analysis was conducted after achieving thermodynamic equilibrium of the system. The same analysis was performed (b) after applying the electric field, $\vec{E}$, and (c) after removing the electric field and reestablishing equilibrium, respectively.

align well along the surface via electrostatic interactions with the PVDF polymer. Consequently, Groups (1) and (3), located above and below the nanoparticles along the z-direction, effectively maintain strong polarization densities because the nanoparticle–PVDF interaction direction is aligned with the direction of the electric field. However, differences between Groups (1) and (3) are observed, with Group (1) exhibiting the highest polarization density after the electric field is removed. This is because the direction of the adsorption force between PVDF and the nanoparticles and the direction of the electric field are the same in Group (1). In addition, the strongly polarized PVDF chains exhibit selective adsorption of polar fillers [52,53]. In other words, fluorine atoms with higher partial charges interact more strongly with Al_2O_3 nanoparticles (see Fig. 1**(d)**). In Group (1), the fluorine atoms of PVDF face the nanoparticle under an applied electric field; hence, the polarization density is maintained at a high level. By contrast, in Group (3), the hydrogen atoms of the chain are forced to face the nanoparticles, and the polarization-locking effect is less pronounced. As a result, Group (1) maintains a higher polarization density than Group (3), particularly in the area immediately adjacent to the surface of the Al_2O_3 nanoparticles, where the electrostatic interaction is the most pronounced. The polarization reinforcement or cancellation characteristics by adsorption with nanofillers and the electric field response exhibited by the PVDF chains are also in good agreement with the results of the MD simulation study performed by Shepelin et al. on a PVDF/CNT interface [54]. By contrast, Group (2), where the interaction vector between the nanoparticles and PVDF chains acts nearly perpendicular to the electric field vector, exhibits a more pronounced springback in the polarization density after the removal of the electric field. In this region, the adsorption force and electric field of the nanoparticles compete with each other. This results in a significant decrease in the overall polarization density in Group (2) compared with those of Groups (1) and (3).

These findings demonstrate that the relative positions and

interaction dynamics between the nanoparticles and PVDF chains play an important role in determining the local polarization behavior. The structural changes in the PVDF chains resulting from this polarization behavior are detailed in the following sections.

3.3. Spatially dependent structural changes of PVDF molecules around Al_2O_3 nanoparticles

The variations in the polarization density observed among the spatial groups of PVDF chains surrounding the Al_2O_3 nanoparticle are attributed to structural changes induced by the external electric field and nanoparticle interactions. To better understand the structural changes, the structural behavior of the PVDF chains was evaluated in terms of two independent factors: (1) the backbone morphology represented by the dihedral angles within the polymer chain and (2) the orientation of the chain itself.

The structural characteristics of the PVDF chains according to their positions relative to the nanoparticles are illustrated in Fig. 5. For the PVDF chains belonging to Group (2), the electric field and nanoparticle interactions are perpendicular to each other. The orthogonal relationship hinders the effective maintenance of the polarized structures of the PVDF chains by reducing the chain alignment and destabilizing the *trans* structures. Furthermore, the spatial constraints imposed by the nanoparticles force the PVDF chains to align perpendicular to the electric field, resulting in the lowest polarization density. By contrast, Groups (1) and (3) exhibit parallel conformations between the electric field and nanoparticle interactions. The PVDF chains with *trans* conformations stably maintain their metastable state owing to the parallel alignment, leading to enhanced polarization properties. However, owing to the difference in the PVDF terminal groups, the structural locking effect imparted by the nanoparticles for Groups (1) and (3) differ in magnitude. Specifically, the best polarization properties are observed in Group (1) owing to the direct interaction of the F atoms with a high partial charge on the nanoparticle surface. Conversely, for chains in Group (3), the H atoms are in closer contact with the nanoparticles; thus, the locking effect is relatively weak. This causes a larger polarization springback when the electric field is removed.

To quantitatively understand the mechanism presented above, the dihedral angles of the PVDF chains, which shifted toward the *gauche*-to-*trans* state under the influence of an electrostatic stimulus, were quantified. The dihedral angles of all C atoms constituting the PVDF backbone were calculated, and the *trans* ratio was determined based on the proportion of angles that fell within the range of 120°–240°. The results showed that both the external electric field and interaction with the Al_2O_3 nanoparticles significantly affected the dihedral angle distribution of the PVDF chains. Under thermodynamic equilibrium without electric field application, the dihedral angles of the PVDF chains exhibited low *trans* ratios uniformly in all spatial regions (Fig. 6**(a)**). However, upon the application of an electric field, the PVDF chains underwent significant structural rearrangement, resulting in an increase in the *trans* ratio (Fig. 6**(b)**). This transformation was attributed to the asymmetric partial charges of the terminal H and F atoms of the PVDF chains, which tended to align in the direction of the electric field. After the electric field was removed, the dihedral angles of the PVDF backbones permanently changed, and the chains remained in a metastable *trans* state (Fig. 6**(c)**).

The interaction of the spherical Al_2O_3 nanoparticles with PVDF also affected the extent of the dihedral angle change. For the final state, in which the application and removal of the electric field were complete, the *trans* ratio in Group (2) decreased by 13.0 %, whereas Groups (1) and (3) exhibited smaller reductions of 11.2 % and 11.5 %, respectively. Such differences were also observed in the polarization density results with the same tendency. The PVDF chains belonging to Groups (1) and (3) interacted with the Al_2O_3 nanoparticles in a direction parallel to the applied electric field, thereby maintaining a higher *trans* ratio. In particular, Group (1) showed a higher *trans* ratio than Group (3) because the adsorption direction of the polar nanoparticles and the orientation direction owing to the electric field were consistent. However, the orthogonal interaction in Group (2) hindered the maintenance of the altered dihedral angles, resulting in an overall lower *trans* ratio compared with that of the other groups.

Subsequently, we focused on the alignment properties of the entire PVDF chain within the microstructures. The orientation behavior of the PVDF chains was analyzed using the orientational order parameter, which is expressed as follows:

$$S = \left\langle \frac{3cos^2\theta - 1}{2} \right\rangle \tag{4}$$

where θ indicates the angle between the PVDF backbone direction and the electric field direction, and the bracket symbol $\langle\ \rangle$ denotes the average over all evaluated cases. The detailed methods for calculating θ within the microstructure are also provided in **Fig. S3** in the Supplementary Data.

For the nanocomposite models that reached thermodynamic equilibrium, the orientational orders of the PVDF chains remained low overall (Fig. 7**(a)**). Nevertheless, the space group analysis showed a distinct pattern that differed from the observed trends in the polarization density and *trans* ratio results. The orientational orders of Groups (1) and (3) within 3 Å from the nanoparticle surface were lower than –0.1, indicating a strong alignment perpendicular to the direction of the electric field. This alignment was mainly driven by the spherical shape of the nanoparticles. In other words, the PVDF chains were attached to the nanoparticle surfaces and coiled along their curvatures.

When an electric field is applied, the PVDF chains are predominantly aligned perpendicular to the field direction, yielding negative orientational orders (Fig. 7**(b)**). Because this alignment is induced by the morphology of the nanoparticles rather than by direct electrostatic interactions with the nanoparticles, no significant differences are observed between Groups (1) and (3). By contrast, Group (2) shows a lower degree of alignment than the other two groups throughout the entire process. This behavior is attributed to the spatial constraints imposed by the nanoparticles, which restrict the macroscopic orientational motion of the PVDF chains near the nanoparticle interface. Although the applied electric field tends to align the PVDF chains perpendicular to their direction, the nanoparticles restrict the mobility of the chains in this

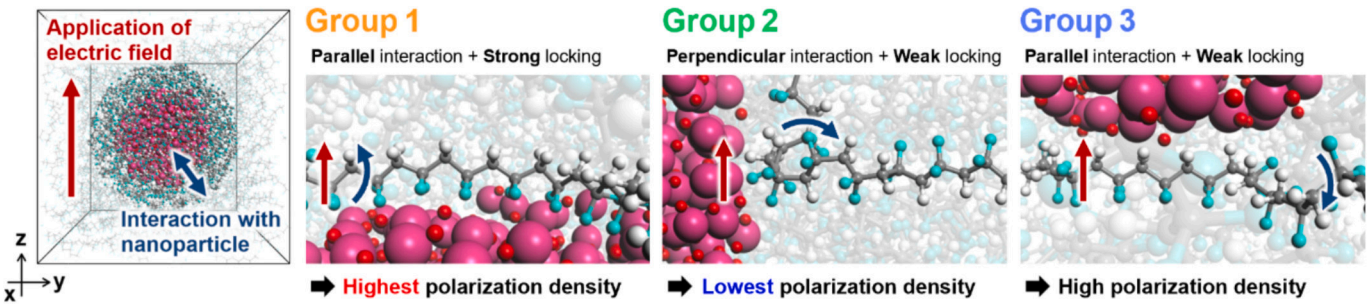


Fig. 5. Principle of differences in polarization characteristics by spatial group around the Al_2O_3 nanoparticle.

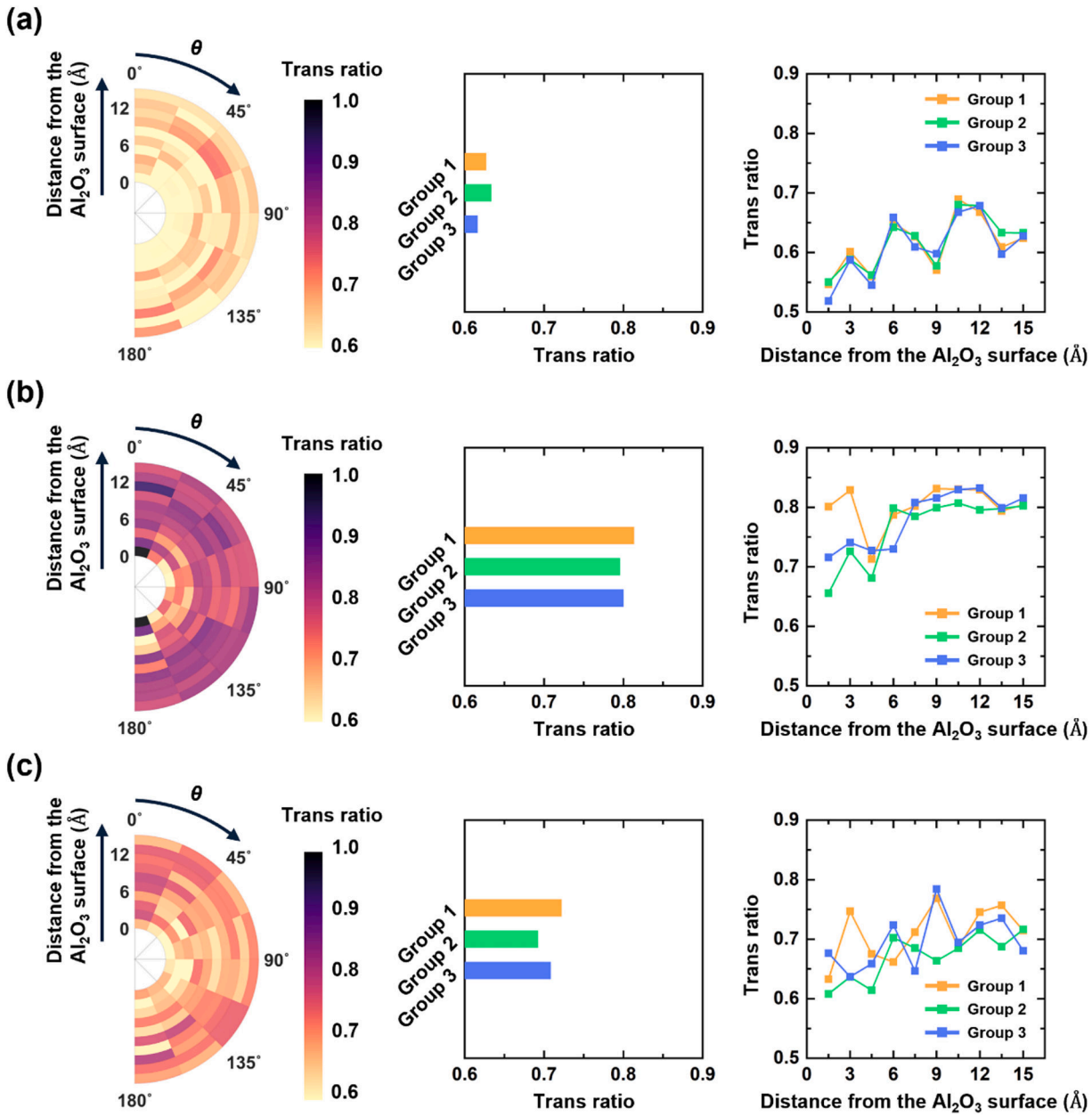


Fig. 6. *Trans* ratio of the dihedral angle of the PVDF backbone surrounding Al_2O_3 nanoparticles. (a) Spatial distribution of the *trans* ratio relative to the Al_2O_3 nanoparticle (left); fraction of the *trans* state occupied by each of the regions defined as Groups (1), 2, and 3 (middle); and radial distribution of the *trans* ratio as a function of the distance from the Al_2O_3 nanoparticle surface (right). The analysis was conducted after achieving thermodynamic equilibrium in the system. The same analysis was performed (b) after applying the electric field, $\vec{E}$, and (c) after removing the electric field and reestablishing equilibrium, respectively.

region. Notably, within the 3 Å region, the orientational order remains close to zero at all stages, indicating that the chains in this region fail to align effectively under the influence of the electric field. Finally, the spatially dependent orientation of the PVDF chains remains significant, even after the removal of the electric field (Fig. 7**(c)**).

In summary, the spatially distinct polarization density of PVDF/Al_2O_3 arises from the following mechanism. First, an external electric field induces unidirectional transformations in the dihedral angle and orientation of the PVDF chains, which are independent of the spatial position and result in permanent polarization changes. Second, the presence of Al_2O_3 nanoparticles causes the PVDF chains to adsorb around the nanoparticles, enabling the polarization of the PVDF chains to remain strong, even after the electric field is removed. The level of residual polarization and its arrangement characteristics in the PVDF chains are determined by the sum of the electric field and adsorption force. In other words, the polarization characteristics depend on the relative positions of the PVDF chains formed with the Al_2O_3 nanoparticles.

4. Conclusions

In this study, we elucidated the molecular origin of the spatial dependence of the polarization density of PVDF-based nanocomposites induced during electrical poling. By analyzing the structural transformations and orientation of the PVDF chains, this study demonstrates how the combined effects of an external electric field and nanoparticle-induced locking generate distinct polarizations across spatial groups.

MD simulations performed under cyclic electric fields showed spatial selectivity in the polarization and subsequent rearrangement characteristics of the PVDF chains around the Al_2O_3 nanoparticles. Depending

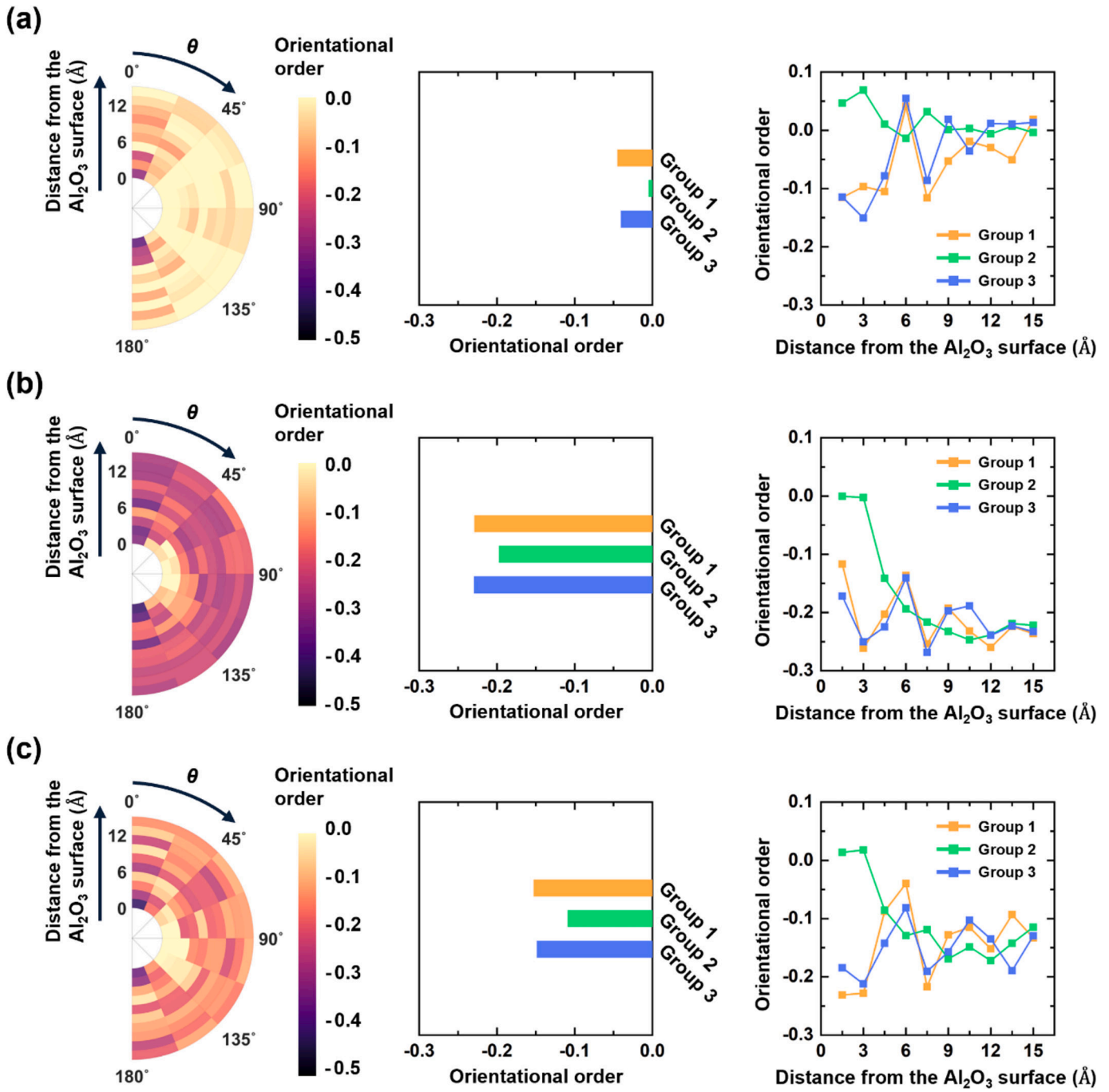


Fig. 7. Orientational order characteristics of the PVDF chains surrounding Al_2O_3 nanoparticles. (a) Spatial distribution of the orientational order relative to the Al_2O_3 nanoparticle (left); average orientational order of PVDF chains occupied by each of the regions defined as Groups (1), 2, and 3 (middle); and radial distribution of the orientational order as a function of the distance from the Al_2O_3 nanoparticle surface (right). The analysis was conducted after achieving thermodynamic equilibrium in the system. The same analysis was performed (b) after applying the electric field, $\vec{E}$, and (c) after removing the electric field and reestablishing equilibrium, respectively.

on the relative positions of the nanoparticles and direction of the applied electric field, the PVDF chains in the matrix region exhibited different structural characteristics and polarization properties. The key to maximizing the polarization of the PVDF chains is to align the direction of the electric field force and the direction of the adsorption force with the nanoparticles such that they are parallel to each other.

These findings highlight the critical role of the nanoparticle morphology and spatial distribution in determining the polarization performance of PVDF-based nanocomposites. The insights gained from this study extend beyond polarization, offering a broader perspective on the influence of nanofiller interactions on the behavior of polymer matrices in various nanocomposite systems. Therefore, this study provides theoretical guidance for designing advanced nanocomposite materials that can leverage synergistic nanoparticle interactions to achieve enhanced mechanical, electrical, and optical properties. Building on the atomistic insights gained in this study, a continuum-based multiscale modeling framework incorporating the spatial distribution of polarization density will be presented in the near future. In modeling the homogenization of interfaces formed by the content and distribution of nanoparticles, the local changes in polarization efficiency obtained

through this work provide important input parameters that can be applied in mesoscopic and macroscopic regimes. Further studies are also underway to characterize the shape and arrangement of nanoparticles to optimize their associated multifunctionality.

CRediT authorship contribution statement

Yeongbin Kim: Writing – original draft, Visualization, Software, Methodology. **Hongdeok Kim:** Writing – review & editing, Methodology. **Joonmyung Choi:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compositesa.2025.109224.

Data availability

Data will be made available on request.

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